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Lecture 11A

Before-Tutorial-Session

Tut09

HIDDEN

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Recall:

$$\langle \underline{x}, \underline{x} \rangle \geq 0$$

Proposition X. The Euclidean inner product obeys the following properties:

- (Positivity) **BLANK 1** for all $\underline{x} \in \mathbb{R}^n$.
- (Definiteness) **BLANK 2**
- (Additivity in first slot) $\langle \underline{x} + \underline{y}, \underline{z} \rangle = \langle \underline{x}, \underline{z} \rangle + \langle \underline{y}, \underline{z} \rangle$ for all $\underline{x}, \underline{y}, \underline{z} \in \mathbb{R}^n$.
- (Homogeneity in first slot) $\langle \lambda \underline{x}, \underline{y} \rangle = \lambda \langle \underline{x}, \underline{y} \rangle$ for all $\underline{x}, \underline{y} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
- (Symmetry) $\langle \underline{x}, \underline{y} \rangle = \langle \underline{y}, \underline{x} \rangle$ for all $\underline{x}, \underline{y} \in \mathbb{R}^n$.

} this is about $\langle \underline{x}, \underline{x} \rangle$

$\langle \underline{x}, \underline{x} \rangle = 0$ iff $\underline{x} = \underline{0}$

↓
dot product

Let's do these questions today. Tut09, Q2+Q7

→ straight forward

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

- (a) If $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle$, then $\mathbf{v} = \mathbf{w}$.
- (b) If $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle$ for all $\mathbf{u} \in \mathcal{V}$, then $\mathbf{v} = \mathbf{w}$.

Question 7. Let $\mathbf{u}_1 = (-1, 2, 2)^T$, $\mathbf{u}_2 = (2, 2, -1)^T$ and $\mathbf{b} = (1, 0, 0)$.

- (a) Show that $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set.
- (b) Find the orthogonal projections $\mathbf{v}_1 \triangleq \text{Proj}_{\mathbf{u}_1}(\mathbf{b})$ and $\mathbf{v}_2 \triangleq \text{Proj}_{\mathbf{u}_2}(\mathbf{b})$.
- (c) Set $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. In what follows, we verify that $\mathbf{v} = \text{Proj}_{\mathcal{U}}(\mathbf{b})$, where $\mathcal{U} = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$. Let $\mathbf{e} = \mathbf{b} - \mathbf{v}$. Show that $\langle \mathbf{e}, \mathbf{u}_i \rangle = 0$ for $i = 1, 2$.
- (d) Let $\mathbf{w}_1 = (1, 0, -2)$ and $\mathbf{w}_2 = (0, 2, 1)$. Hence, $\mathcal{U} = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$ (you need not prove this). Find the orthogonal projections $\mathbf{x}_1 \triangleq \text{Proj}_{\mathbf{w}_1}(\mathbf{b})$ and $\mathbf{x}_2 \triangleq \text{Proj}_{\mathbf{w}_2}(\mathbf{b})$. However, we observe that $\mathbf{v} \neq \mathbf{x}_1 + \mathbf{x}_2$. Explain $\text{Proj}_{\mathcal{U}}(\mathbf{b})$ is not $\text{Proj}_{\mathbf{w}_1}(\mathbf{b}) + \text{Proj}_{\mathbf{w}_2}(\mathbf{b})$.

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(a) If $\langle u, v \rangle = \langle u, w \rangle$, then $v = w$.

correct

Albert's Solution:

Consider $n = 2$. Set $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 0 \\ 1048576 \end{bmatrix}$.

So, $\langle u, v \rangle = \langle u, w \rangle = 0$.

But $v \neq w$.

Therefore, the statement is not true.

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Counterexample to
show statement is false
→ one counterexample
is sufficient

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Set $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 0 \\ 1048576 \end{bmatrix}$.

Is it true that $\langle u, v \rangle = \langle u, w \rangle$ for all u ? $\Rightarrow$ Not true

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \langle u, v \rangle = \langle u, w \rangle = 0$$

$$\begin{aligned} u = \begin{bmatrix} 0 \\ 1 \end{bmatrix} &\Rightarrow \langle u, v \rangle = 1 \\ &\langle u, w \rangle = 1048576 \end{aligned} \left. \vphantom{\begin{aligned} u = \begin{bmatrix} 0 \\ 1 \end{bmatrix} &\Rightarrow \langle u, v \rangle = 1 \\ &\langle u, w \rangle = 1048576 \end{aligned}} \right\} \langle u, v \rangle \neq \langle u, w \rangle$$

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

Let's prove Q2(b) using Proposition X.

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$\langle u, v \rangle = \langle u, w \rangle$ implies that $\langle u, v \rangle - \langle u, w \rangle =$ **BLANK**

0 → number!

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Let's prove Q2(b) using Proposition X.

$$\langle u, v \rangle - \langle u, w \rangle = \langle v, u \rangle - \langle w, u \rangle$$

Why?

Symmetric property

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Let's prove Q2(b) using Proposition X.

$$\langle v, u \rangle - \langle w, u \rangle = \langle v, u \rangle + \langle -w, u \rangle$$

Why?

Homogeneous in first slot

1st slot 2nd slot
↓ ↓
< , >

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Let's prove Q2(b) using Proposition X.

$$\langle v, u \rangle + \langle -w, u \rangle = \langle v - w, u \rangle$$

Why?

Additive in first slot

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Let's prove Q2(b) using Proposition X.

$$\langle v - w, v - w \rangle = 0$$

Why?

$$\begin{aligned} & \langle \underline{u}, \underline{v} \rangle = \langle \underline{u}, \underline{w} \rangle \\ \Rightarrow \dots & \Rightarrow \langle \underline{v} - \underline{w}, \underline{u} \rangle = 0 \quad \left. \begin{array}{l} \text{for all } \underline{u} \\ \text{choose any } \underline{u} \text{ to put here} \\ \text{in this case, choose } \underline{u} = \underline{v} - \underline{w} \end{array} \right\} \end{aligned}$$

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

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Let's prove Q2(b) using Proposition X.

$\langle v - w, v - w \rangle = 0$ implies that $v - w = \mathbf{0}$ ↗ vector

Why?

Definiteness

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let u , v , and w belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(b) If $\langle u, v \rangle = \langle u, w \rangle$ for all $u \in \mathcal{V}$, then $v = w$.

Proof

$$\langle u, v \rangle = \langle u, w \rangle$$

$$\Rightarrow 0 = \langle u, v \rangle - \langle u, w \rangle$$

$$= \langle u, v - w \rangle$$

$$\text{Choose } u = v - w \text{ and so } \langle v - w, v - w \rangle = 0$$

$$\text{By definition } \Rightarrow v - w = 0$$

$$\therefore v = w$$

Tutorial 9 Q1

→ show homogeneous & additive in 2nd slot

Question 7. Let $\mathbf{u}_1 = (-1, 2, 2)^T$, $\mathbf{u}_2 = (2, 2, -1)^T$ and $\mathbf{b} = (1, 0, 0)$.

- (a) Show that $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set.
- (b) Find the orthogonal projections $\mathbf{v}_1 \triangleq \text{Proj}_{\mathbf{u}_1}(\mathbf{b})$ and $\mathbf{v}_2 \triangleq \text{Proj}_{\mathbf{u}_2}(\mathbf{b})$.
- (c) Set $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. In what follows, we verify that $\mathbf{v} = \text{Proj}_{\mathcal{U}}(\mathbf{b})$, where $\mathcal{U} = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$. Let $\mathbf{e} = \mathbf{b} - \mathbf{v}$. Show that $\langle \mathbf{e}, \mathbf{u}_i \rangle = 0$ for $i = 1, 2$.
- (d) Let $\mathbf{w}_1 = (1, 0, -2)$ and $\mathbf{w}_2 = (0, 2, 1)$. Hence, $\mathcal{U} = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$ (you need not prove this). Find the orthogonal projections $\mathbf{x}_1 \triangleq \text{Proj}_{\mathbf{w}_1}(\mathbf{b})$ and $\mathbf{x}_2 \triangleq \text{Proj}_{\mathbf{w}_2}(\mathbf{b})$. However, we observe that $\mathbf{v} \neq \mathbf{x}_1 + \mathbf{x}_2$. Explain $\text{Proj}_{\mathcal{U}}(\mathbf{b})$ is not $\text{Proj}_{\mathbf{w}_1}(\mathbf{b}) + \text{Proj}_{\mathbf{w}_2}(\mathbf{b})$.

In what follows, I will try to motivate the formula.

Detailed computations will be omitted.

Question 7. Let $u_1 = (-1, 2, 2)^T$, $u_2 = (2, 2, -1)^T$ and $b = (1, 0, 0)$.

(a) Show that $S = \{u_1, u_2\}$ is an orthogonal set.

Our Task: Show that $\langle u_1, u_2 \rangle =$ **BLANK**

$\rightarrow$ number! not vector.

— S orthogonal means that $\langle u, v \rangle = 0$

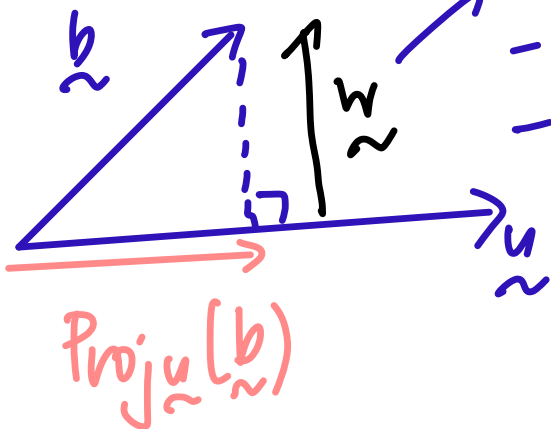
for any pair of distinct u, v .

Question 7. Let $u_1 = (-1, 2, 2)^T$, $u_2 = (2, 2, -1)^T$ and $b = (1, 0, 0)$.

(b) Find the orthogonal projections $v_1 \triangleq \text{Proj}_{u_1}(b)$ and $v_2 \triangleq \text{Proj}_{u_2}(b)$.

Our Task: We use the formula: $\text{Proj}_u(b) =$ **BLANK**.

try both variations
 - compute $\tilde{u} = b - \text{Proj}_u(b)$
 - check $\langle \tilde{u}, u \rangle = 0$



$$\frac{\langle \cdot \rangle}{\langle \cdot \rangle} \rightarrow \frac{\langle b, \tilde{u} \rangle}{\langle \tilde{u}, \tilde{u} \rangle} \text{ or } \frac{\langle \tilde{u}, \tilde{u} \rangle}{\langle b, \tilde{u} \rangle}$$

↓
correct answer

↓
A vector! $\Rightarrow$ number $\times$ vector
 ↓
should be $\tilde{u}$

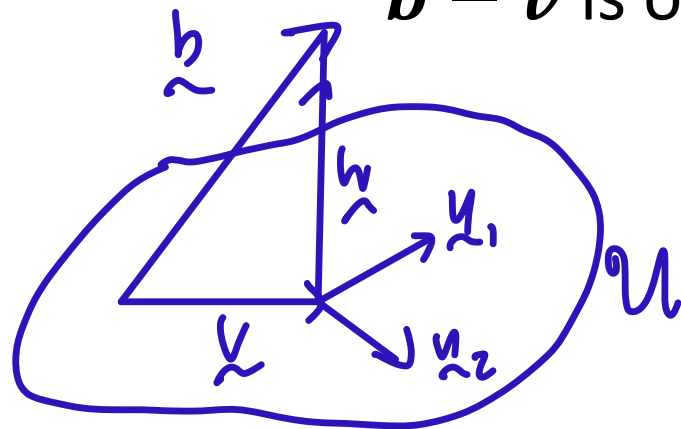
Question 7. Let $u_1 = (-1, 2, 2)^T$, $u_2 = (2, 2, -1)^T$ and $b = (1, 0, 0)$.

(c) Set $v = v_1 + v_2$. In what follows, we verify that $v = \text{Proj}_{\mathcal{U}}(b)$, where $\mathcal{U} = \text{span}(u_1, u_2)$. Let $e = b - v$. Show that $\langle e, u_i \rangle = 0$ for $i = 1, 2$.

Recall: If $v = \text{Proj}_{\mathcal{U}}(b)$, then

$b - v$ is orthogonal to u for

BLANK



$\underline{w} = \underline{b} - \underline{v}$ is orthogonal to $\underline{u}$ for all $\underline{u}$ in $\mathcal{U}$.

i.e. $\langle \underline{b} - \underline{v}, \underline{u} \rangle = 0$ for all $\underline{u}$ in $\mathcal{U}$.

check only 2 vectors u_1, u_2

$\Rightarrow$ because $[u_1, u_2]$ is a basis for $\mathcal{U}$,

$\langle \underline{b} - \underline{v}, \underline{u} \rangle = 0$ for $\underline{u}$ in $\mathcal{U}$ if and only if

$\langle \underline{b} - \underline{v}, u_1 \rangle = \langle \underline{b} - \underline{v}, u_2 \rangle = 0$

ALL $\underline{u} \in \mathcal{U}$.

Question 7. Let $\mathbf{u}_1 = (-1, 2, 2)^T$, $\mathbf{u}_2 = (2, 2, -1)^T$ and $\mathbf{b} = (1, 0, 0)$.

- (c) Set $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. In what follows, we verify that $\mathbf{v} = \text{Proj}_{\mathcal{U}}(\mathbf{b})$, where $\mathcal{U} = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$. Let $\mathbf{e} = \mathbf{b} - \mathbf{v}$. Show that $\langle \mathbf{e}, \mathbf{u}_i \rangle = 0$ for $i = 1, 2$.

Recall: If $\mathbf{v} = \text{Proj}_{\mathcal{U}}(\mathbf{b})$, then

$\mathbf{b} - \mathbf{v}$ is orthogonal to $\mathbf{u}$ for **BLANK**

Question 7. Let $u_1 = (-1, 2, 2)^T$, $u_2 = (2, 2, -1)^T$ and $b = (1, 0, 0)$.

Recall: If $[u_1, \dots, u_k]$ ^{is} ~~be~~ **BLANK** basis for $\mathcal{U}$,
then
↓ an orthogonal

$$\text{Proj}_{\mathcal{U}}(b) = \frac{\langle b, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle b, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 + \dots + \frac{\langle b, u_k \rangle}{\langle u_k, u_k \rangle} u_k.$$

7(d): Just check that the formula is NOT correct when $[u_1, u_2]$ is NOT orthogonal.

(d) Let $w_1 = (1, 0, -2)$ and $w_2 = (0, 2, 1)$. Hence, $\mathcal{U} = \text{span}(w_1, w_2)$ (you need not prove this).

Find the orthogonal projections $x_1 \triangleq \text{Proj}_{w_1}(b)$ and $x_2 \triangleq \text{Proj}_{w_2}(b)$. However, we observe that $v \neq x_1 + x_2$. Explain why $\text{Proj}_{\mathcal{U}}(b)$ is not $\text{Proj}_{w_1}(b) + \text{Proj}_{w_2}(b)$.